**Definition** (Diagramatic invariant). *$I : knot \to S$ with $S$ set such that $I(K_1) = I(K_2)$ if $K_1$ and $K_2$ are isotopic.*

**Example.** *The crossing at a cross $\varepsilon(c) \in \{+1, -1\}$. The linking number of a link $L = K_1 \cup K_2$ is an invariant defined as*

$$lk(K_1, K_2) = \frac{1}{2} \sum_{c \in K_1 \cap K_2} \varepsilon(c).$$

**Definition** (Fox $p$-coloring). *Let $D$ a knot diagram with arcs $x_1, \dots, x_m$. A Fox $p$-coloring of $D$ is an assignment of colors*

$$x_i \mapsto a_i \in \mathbb{Z}_p$$

*such that at each crossing, we have $a, c$ underarcs and $b$ overarc, then $2b \equiv a + c \pmod{p}$.*

*We have a homogeneous system of equations $C_D x = 0$*

**Proposition.** *The dimension of*

$$Col_p(D) = \ker(C_D)$$

*(hence, the number of Fox p-colorings) is a knot invariant.*

**Definition** (Knot group). *Let $K$ be a knot. The knot group of $K$ is $\pi_1(S^3 \setminus K)$.*

**Definition.** *$N(K)$ tubular nbh of $K$.*

**Definition** (Meridian and longitude). *They generate $\pi_1(\partial N(K)) \cong \mathbb{Z}^2$.*

**Theorem** (Wirtinger presentation). *$g_i$ generator for each arc $x_i$. Then the knot group has presentation*

$$G_K = \langle g_1, \dots, g_m \mid r_1, \dots, r_n \rangle$$

*where $r_j : g_k = g_i g_j g_i^{-1}$ are wirtinger relations at each crossing.*

**Definition.** *$C_D^{i,j}$ deleting the $i$-th row and $j$-th column of $C_D$. Then*

$$\det(K) = \left| \det(C_D^{i,j}) \right| = |\Delta_K(-1)| = |H_1(\Sigma_2(K))|$$

**Example.** *Trifoil*

$$C_D = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

**Theorem.** *$K$ admits a non-trivial Fox $p$-coloring if and only if $p \mid \det(K)$.*

**Definition** (Writhe). *Not a knot invariant (fails Redemeister I)*

$$w(D) = \sum_c \varepsilon(c)$$

**Definition** (Framed knot). *A knot $K$ equiped with $v(x)$ a normal vector field. Self-linking number*

$$sl(K) = lk(K, K_v = \{x + \varepsilon v(x) : x \in K\})$$

# 1 Seifert and Alexander

**Definition** (Seifert surface). *For a knot $K$, a Seifert surface is a compact connected orientable surface $F$ such that $\partial F = K$.*

**Proposition.** *Seifert's algorithm: TODO*

**Definition** (Euler characteristic).

$$\chi(F) = s(D) - c(D) = 1 - 2g(F)$$

*where $s(D)$ is the number of Seifert circles and $c(D)$ is the number of crossings.*

**Proposition.** *$K_1 \# K_2$ induces $F_1 \natural F_2$ and*

$$g(F_1 \natural F_2) = g(F_1) + g(F_2), \quad g(K_1 \# K_2) = g(K_1) + g(K_2).$$

**Proposition.** *$F$ Seifert surface of $K$ implies $H_1(F, \mathbb{Z}) \cong \mathbb{Z}^{2g(F)}$*

**Definition** (Seifert matrix)**.** *Basis $\alpha_1, \dots, \alpha_{2g}$ of $H_1(F, \mathbb{Z})$, the Seifert matrix $V$ is defined by*

$$V_{ij} = lk(\alpha_i, \alpha_j^+)$$

*where $\alpha_j^+$ is a push-off of $\alpha_j$ in the positive normal direction of $F$.*

**Definition** (Algebraic intersection number)**.** *$F$ oriented, $a, b$ oriented simple closed curves on $F$. Then*

$$a \cdot b = \sum_{p \in a \cap b} \varepsilon_p(a, b), \quad \varepsilon_p(a, b) = \begin{cases} +1 & \textit{if } (a, b) \textit{ is + oriented at } p \\ -1 & \textit{if } (a, b) \textit{ is - oriented at } p \end{cases}$$

*we also see that $\alpha_i \cdot \alpha_j = V_{ij} - V_{ji}$*

**Definition** (Alexander polynomial)**.** *$V$ Seifert matrix for $K$.*

$$\Delta_K(t) \coloneqq \det(V - tV^T) \in \mathbb{Z}[t, t^{-1}], \quad \Delta_K(t) = \Delta_K(t^{-1}), \quad \Delta_K(1) = 1.$$

**Example.** *Trefoil $V = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, $\Delta_K(t) = t^2 - t + 1 = t - 1 + t^{-1}$*